

MACSE622 – Data Engineering and Visualization LAB

Register Number	25MCB1010	Subject Code	MACSE622
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Programme	M.Tech	Slot	L33+L34
Course	CSE (Big Data Analytics)	Semester	2
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Problem Statement:

Consider a publicly available dataset (which has both numerical and categorical values) and draw the following plots(with title, legend, xlabel, ylabel etc). For each of the plots, draw a inference from the visualization of the plot

Line Plot, Scatter Plot, Bar plot, Histogram, Box Plot, Pie Chart, Pat plot, Sub plot

Program & Output

1. Line Plot:

```
plt.figure()

df.groupby('HouseAge')['MedHouseVal'].mean().plot()

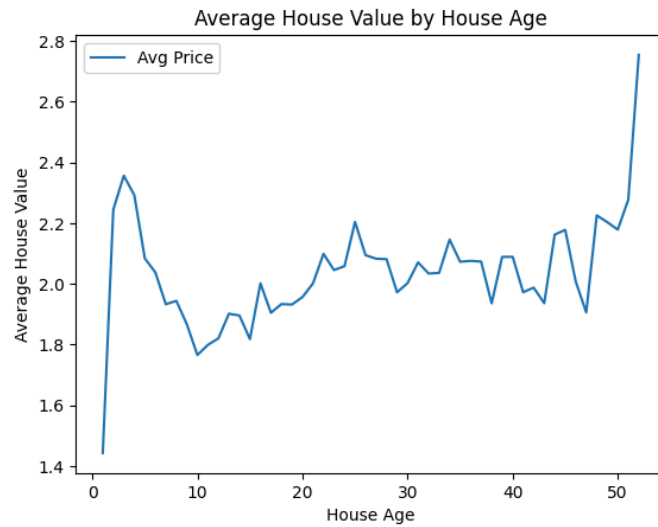
plt.title("Average House Value by House Age")

plt.xlabel("House Age")

plt.ylabel("Average House Value")

plt.legend(["Avg Price"])

plt.show()
```



House value increases with house age, until 50 it increases rapidly

2. Scatter Plot

```
plt.figure()
plt.scatter(df['MedInc'], df['MedHouseVal'], alpha=0.3)
plt.title("Median Income vs House Value")
plt.xlabel("Median Income")
plt.ylabel("House Value")
plt.show()
```



Strong positive correlation between median income and house value

3. Bar Graph

```
plt.figure()

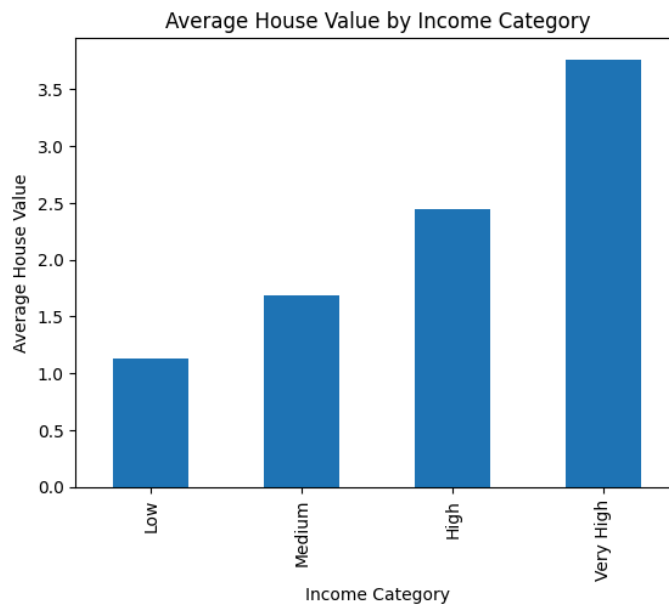
df.groupby('Income_Category')['MedHouseVal'].mean().plot(kind='bar')

plt.title("Average House Value by Income Category")

plt.xlabel("Income Category")

plt.ylabel("Average House Value")

plt.show()
```



Higher income category have significantly higher house prices

4. Histogram

```
plt.figure()

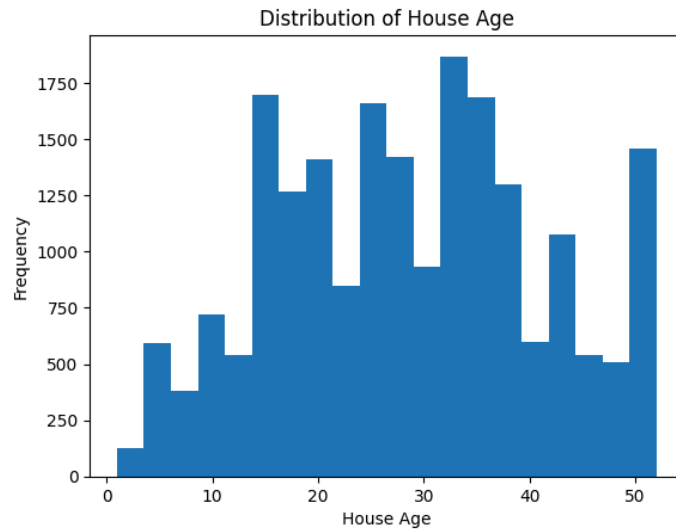
plt.hist(df['HouseAge'], bins=20)

plt.title("Distribution of House Age")

plt.xlabel("House Age")

plt.ylabel("Frequency")

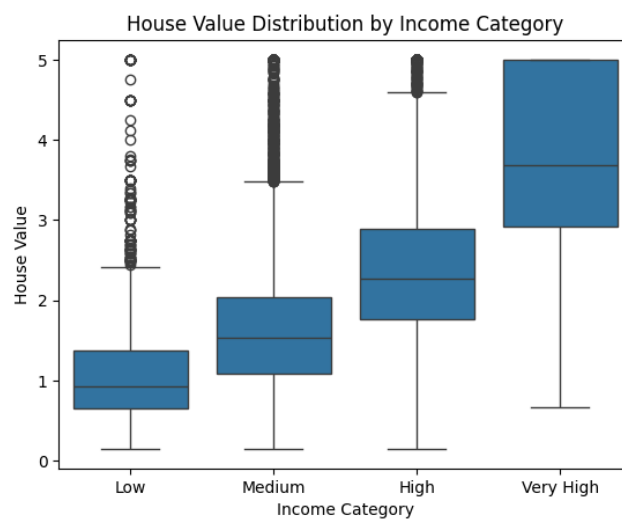
plt.show()
```



Most houses are average aged, very few very old houses

5. Box Plot

```
plt.figure()
sns.boxplot(x='Income_Category', y='MedHouseVal', data=df)
plt.title("House Value Distribution by Income Category")
plt.xlabel("Income Category")
plt.ylabel("House Value")
plt.show()
```



Higher income category show higher median and more spread in prices

6. Pie Chart

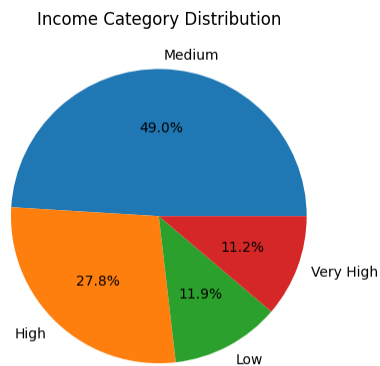
```
plt.figure()

df['Income_Category'].value_counts().plot(kind='pie', autopct='%1.1f%%')

plt.title("Income Category Distribution")

plt.ylabel("")

plt.show()
```

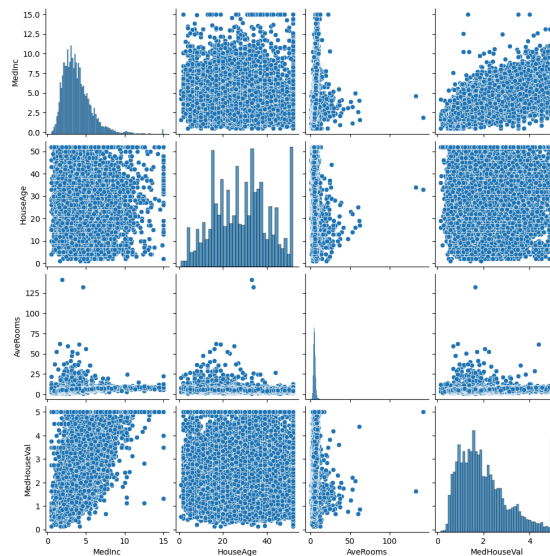


Majority of houses fall under Medium income category

7. Pair Plot

```
sns.pairplot(df[['MedInc', 'HouseAge', 'AveRooms', 'MedHouseVal']])

plt.show()
```



Income is the strongest predictor of house value

8. Sub Plots

```
fig, axes = plt.subplots(2,2)
```

#Histogram

```
axes[0,0].hist(df['HouseAge'], bins=20)
```

```
axes[0,0].set_title("House Age Distribution")
```

#Scatter Plot

```
axes[0,1].scatter(df['MedInc'], df['MedHouseVal'], alpha=0.3)
```

```
axes[0,1].set_title("Income vs House Value")
```

#Bar Plot

```
df.groupby('Income_Category')['MedHouseVal'].mean().plot(kind='bar', ax=axes[1,0])
```

```
axes[1,0].set_title("Avg Price by Income Category")
```

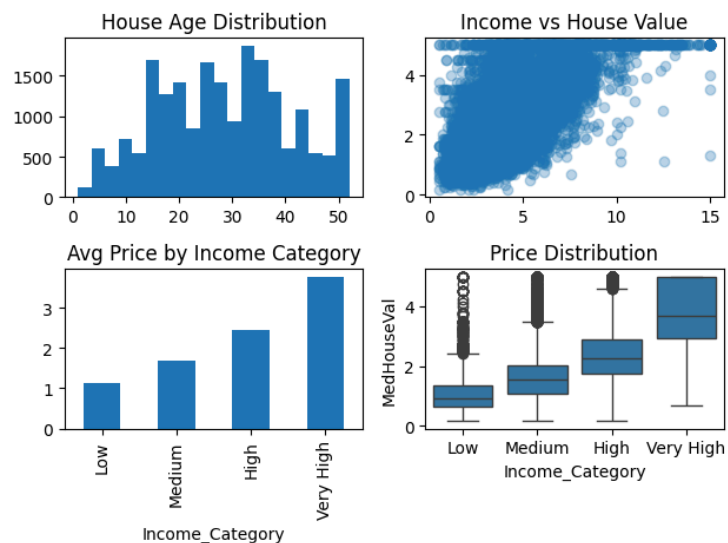
#Box Plot

```
sns.boxplot(x='Income_Category', y='MedHouseVal', data=df, ax=axes[1,1])
```

```
axes[1,1].set_title("Price Distribution")
```

```
plt.tight_layout()
```

```
plt.show()
```



combined view shows income affects pricing more than age